Super-Resolution and Restoration of Images

Leveraging IAPR TC-12 Dataset with Pixel-wise and Perceptual Losses

Low resolution

Inference result

Original High resolution

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The Challenge: Enhancing Low-Quality Images

Image Super-Resolution & Restoration aims to increase resolution (sharpness, detail) and remove degradations (blur, noise). The process is to take as input a low quality image and to have as output a picture with enhanced quality (Image-To-Image).

This kind of task is useful in different scenario: medical images, security camera, satellite imagery and digital media.

Training Data:
IAPR TC - 12 dataset

The original datas are treated as High-Resolution (HR). From these I synthetically created the Low-Resolution (LR) counterparts.

Different degradation were applied:

- Downsample 2x
- Blur
- Noise

The entire process is managed by a *tf.data* pipeline.

Original picture

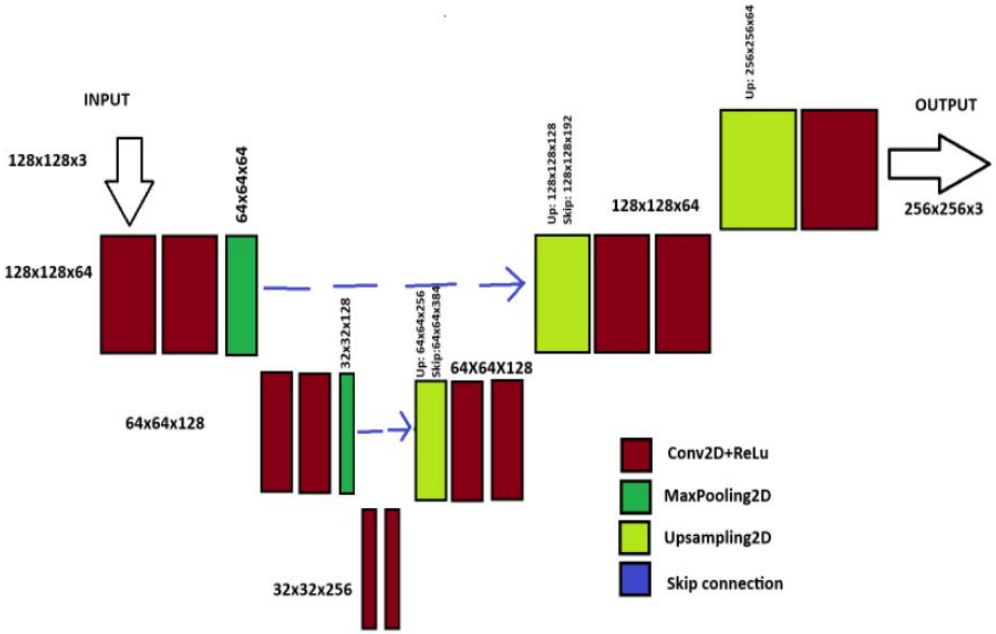
HR patch 256x256

LR patch 128x128

Model Architecture 1: Base U-Net

A U-Net is an Encoder-Decoder architecture with skip connections: they help to recovers spatial information lost during downsampling. The base model take has input 128x128x3 image and has:

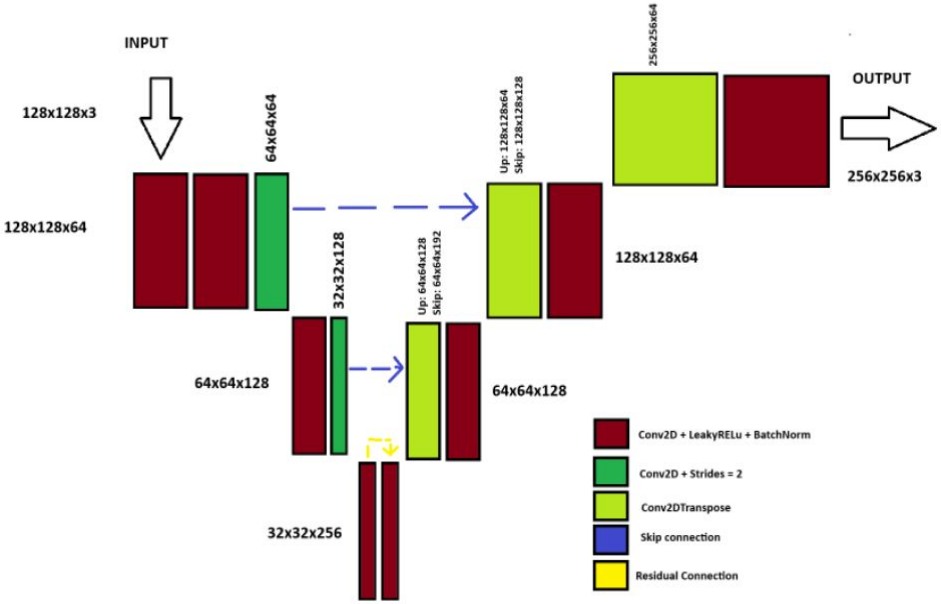
- Encoder: Conv2d(ReLu) -> Maxpooling2D
- Bottleneck: Conv2D (ReLu)
- Decoder: UpSampling2D (bilinear) -> Concatenate (skip connection) -> Conv2D (ReLu)
- Output: Conv2D (Sigmoid) to produce 3-channel RGB image (256x256)



Model architecture 2: Advance U-Net

The second U-Net model I build contains some architecture improvement:

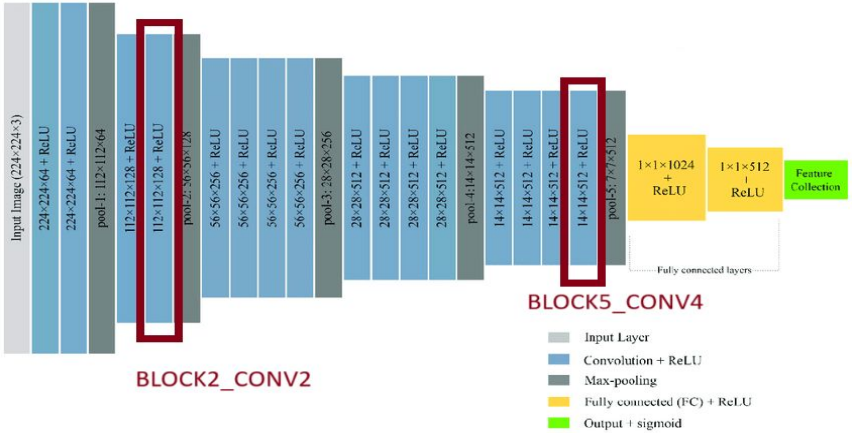
- LeakyReLu: prevent “dying ReLu” problem;
- Strided Convolution: learnable downsampling operation;
- Conv2DTranspose: learnable upsampling operation;
- BatchNormalization: stabilize training and accelerate convergence;
- Residual Skip Connection: aids gradient flow in deeper networks.



Loss Function: Pixel-wise vs Perceptual

- Mean Absolute Error (MAE): Compare pixel values and can lead to blurry results
- Perceptual Loss: Use pre-trained VGG19 (with frozen weight) to extract features. Capture high-level structure
- Combined Loss:

Total Loss = MAE Loss + α * Perceptual Loss



VGG19 representation.
In red the Feature Extraction point.

Training & Evaluation

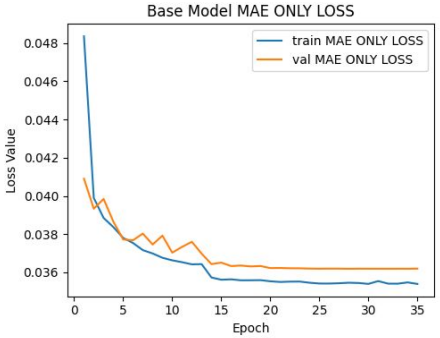
Training:

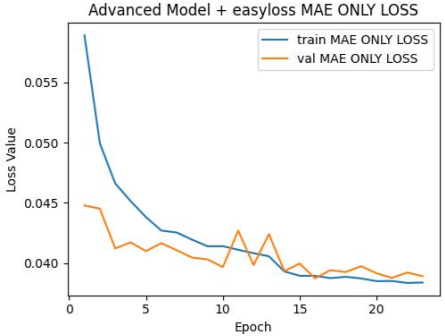
- Optimizer: Adam (lr=5e-4)
- Epochs: 50
- Batch size: 8
- CallBacks:
  - ModelCheckpoint
  - Earlystopping
  - ReduceLRonPlateau (min lr=1e-6)
- Metrics:
  - Loss Function
  - PSNR (Peak Signal-to-Noise)
  - SSIM (Structural Similarity Index)

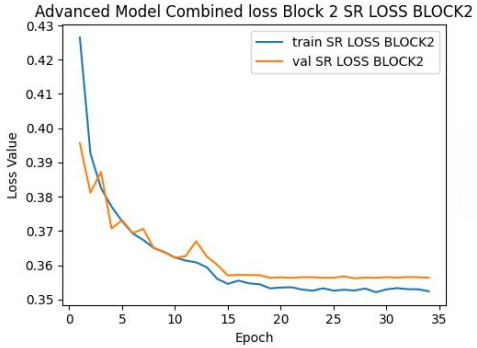
Experiment Conducted:

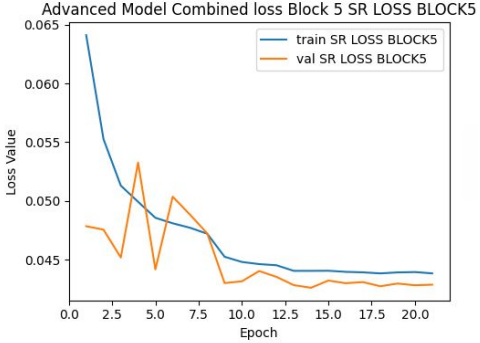
- Base U-Net + MAE Loss
- Adv U-Net + MAE Loss
- Adv U-Net + Combined Loss (block2_Conv2) Alpha = 0.005
- Adv U-Net + Combined Loss (Block5_Conv4) Alpha = 0.005
- Adv U-Net + Combined Loss (Block5_Conv4) Alpha = 0.02

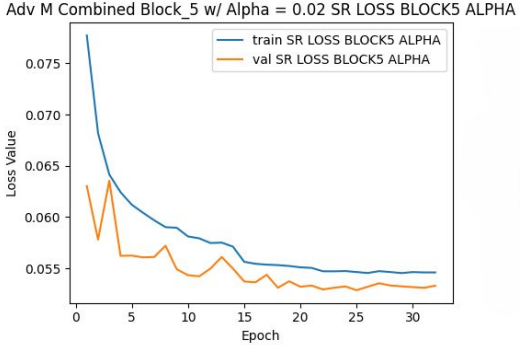
Training Plots









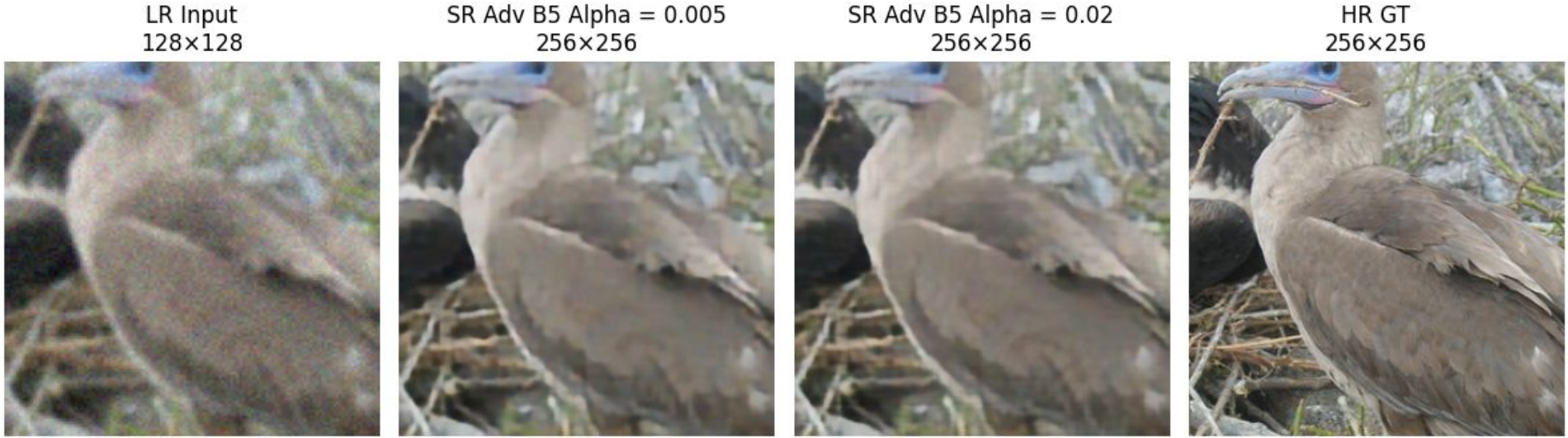


Performance Evaluation: Quantitative Results

Model Conf	Test Loss	MAE	PSNR	SSIM	Perceptual
Base + MAE	0.0364	0.0364	25.7761	0.6990	-
Adv + MAE	0.0389	0.0389	25.4123	0.6793	-
Adv + Comb (Block 2)	0.3582	0.0415	24.9760	0.6467	63.3269
Adv + Comb (Block 5)	0.0428	0.0389	25.3774	0.6769	0.7761
Adv + Comb (Block 5) Alpha = 0.02	0.0531	0.0387	25.4455	0.6817	0.7185

Performance Evaluation: Qualitative Results

Adv U-Net: Alpha = 0.005 vs 0.02



Possible Future Work:

- Focussing only on one task: Super-Resolution or Restoration;
- Use a different dataset where each GT picture have a MUCH higher resolution;
- Explore Generative Adversarial Network (GAN) based losses.

Extra: Trials & Errors

- Use the “correct” GPU;
- Select a better learning rate according to the epochs;
- Complex models are NOT ALWAYS better than simple ones.
- Double check the model before launch the train (!).

Thank You for your attention!